DISCRETE MATHEMATICS & GRAPH THEORY

Module 2

Properties of Integers · Fundamental Principles of Counting

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Lecture Notes

1 Mathematical Induction

The method of mathematical induction is based on a principle called the *Induction Principle*. This principle, in turn, is based on another principle known as the *Well-Ordering Principle*.

Note: $\mathbb{Z}^+$ denotes the set of all positive integers.

Well-Ordering Principle QP

"Every non-empty subset of $\mathbb{Z}^+$ contains a smallest (least) element."

(OR)

"The set of all positive integers is well ordered."

Induction Principle (Principle of Mathematical Induction) QP

Let $S(n)$ denote an open statement that involves a positive integer n. Suppose that the following conditions hold:

(i) $S(1)$ is true.

(ii) Whenever $S(k)$ is true for some particular, arbitrarily chosen $k \in \mathbb{Z}^+$, then $S(k+1)$ is true.

Then $S(n)$ is true for all $n \in \mathbb{Z}^+$.

Method of Mathematical Induction

Suppose we wish to prove that a certain statement $S(n)$ is true $\forall n \geq 1$. The method of proving this using the principle of induction is called the *method of mathematical induction*. There are two steps in this method:

1. **Basic step:** Verify that $S(n)$ is true for $n = 1$.
2. **Induction step:** Assuming $S(k)$ is true, where $k \geq 1$, show that $S(k+1)$ is true.

Note:

1. Without the basic step, one cannot write the proof by induction.
2. Suppose we are required to prove a statement $S(n)$ for $n \geq n_0$, where n_0 is an integer. In such a situation, verifying that $S(n_0)$ is true is the basic step, and in the induction step we take $k \geq n_0$.

Alternate Form of the Induction Principle

Let $S(n)$ denote a statement that involves a positive integer n. Suppose that

(i) for some positive integers $n_0, n_0+1, n_0+2, \ldots, n_0+p$, the statements $S(n_0), S(n_0+1), \ldots, S(n_0+p)$ are true, and

(ii) if $S(n_0), S(n_0+1), \ldots, S(k)$ are true, where $k \geq n_0+p$, then $S(k+1)$ is true.

Then $S(n)$ is true $\forall n \geq n_0$.

1.1 Problems

Problem 1. Prove by mathematical induction that, $\forall n \geq 1$,

$$1+2+3+\cdots+n = \tfrac{1}{2}n(n+1).$$

Solution. Let $S(n): \ 1+2+3+\cdots+n = \frac{1}{2}n(n+1)$.

Basic step: $S(1)$ is the statement $1 = \frac{1}{2}\cdot 1\cdot(2)$, which is true. Thus $S(n)$ is verified for $n = 1$.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., assume

$$S(k): \ 1+2+3+\cdots+k = \tfrac{1}{2}k(k+1).$$

Adding $(k+1)$ to both sides, we get

$$\begin{aligned} 1+2+3+\cdots+k+(k+1) &= \tfrac{1}{2}k(k+1)+(k+1) \\ &= \tfrac{1}{2}(k+1)[k+2] \\ &= \tfrac{1}{2}(k+1)(k+2). \end{aligned}$$

This is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 2. QP Prove by mathematical induction that

$$1^2+3^2+5^2+\cdots+(2n-1)^2 = \tfrac{1}{3}n(2n-1)(2n+1) \qquad \forall n \geq 1.$$

Solution. Let $S(n): \ 1^2+3^2+5^2+\cdots+(2n-1)^2 = \tfrac{1}{3}n(2n-1)(2n+1)$.

Basic step: $S(1): \ 1^2 = \tfrac{1}{3}(1)(1)(3)$, which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e.,

$$1^2+3^2+5^2+\cdots+(2k-1)^2 = \tfrac{1}{3}k(2k-1)(2k+1).$$

Adding $(2k+1)^2$ on both sides $\big($note: $[2(k+1)-1]^2 = (2k+1)^2\big)$:

$$\begin{aligned} 1^2+3^2+\cdots+(2k-1)^2+(2k+1)^2 &= \tfrac{1}{3}k(2k-1)(2k+1)+(2k+1)^2 \\ &= \tfrac{1}{3}(2k+1)[k(2k-1)+3(2k+1)] \\ &= \tfrac{1}{3}(2k+1)[2k^2-k+6k+3] \\ &= \tfrac{1}{3}(2k+1)(2k^2+5k+3) \\ &= \tfrac{1}{3}(k+1)(2k+1)(2k+3). \end{aligned}$$

This is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 3. Prove by mathematical induction that

$$1\cdot 3+2\cdot 4+3\cdot 5+\cdots+n(n+2) = \frac{n(n+1)(2n+7)}{6}.$$

Solution. Let $S(n)$ denote the given statement.

Basic step: $S(1): \ 1\cdot 3 = \dfrac{1\cdot 2\cdot 9}{6} = 3$, which is true.

Induction step: Assume that $S(k)$ is true, where $k \geq 1$, i.e.,

$$1\cdot 3+2\cdot 4+3\cdot 5+\cdots+k(k+2) = \frac{k(k+1)(2k+7)}{6}.$$

Adding $(k+1)(k+3)$ on both sides, we get

$$\begin{aligned} 1\cdot 3+2\cdot 4+\cdots+k(k+2)+(k+1)(k+3) &= \frac{k(k+1)(2k+7)}{6}+(k+1)(k+3) \\ &= \frac{(k+1)}{6}[k(2k+7)+6(k+3)] \\ &= \frac{(k+1)}{6}[2k^2+13k+18] \\ &= \frac{(k+1)(k+2)(2k+9)}{6}. \end{aligned}$$

This is the statement $S(k+1)$, since $2(k+1)+7 = 2k+9$. So $S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 4. QP Prove by mathematical induction that, for every positive integer n, 5 divides $n^5 - n$.

Solution.

> **Note:** "5 divides 30" means $\frac{30}{5}$ is an integer, i.e., 30 is a multiple of 5: $30 = 5 \times m$ for some positive integer m (here $m = 6$).

Let $S(n)$: 5 divides $n^5 - n$.

Basic step: $S(1)$: 5 divides $1^5 - 1 = 0$ (i.e., $\frac{0}{5} = 0$), which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., 5 divides $k^5 - k$. So $k^5 - k$ is a multiple of 5:

$$\boxed{k^5 - k = 5m} \quad \text{for some integer } m.$$

Now

$$\begin{aligned}(k+1)^5 - (k+1) &= (k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1) - (k+1) \\ &= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k) \\ &= 5m + 5(k^4 + 2k^3 + 2k^2 + k) \\ &= 5\,(m + k^4 + 2k^3 + 2k^2 + k).\end{aligned}$$

$\Rightarrow (k+1)^5 - (k+1)$ is a multiple of 5, i.e., 5 divides $(k+1)^5 - (k+1)$. This is exactly the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, for $k \geq 1$. Therefore, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 5. QP Prove that $4n < (n^2 - 7)$ for all positive integers $n \geq 6$.

Solution. Let $S(n)$: $4n < (n^2 - 7)$. We prove that $S(n)$ is true for all positive integers $n \geq n_0$, where $n_0 = 6$.

Basic step: $S(6)$: $(4 \times 6) < (6^2 - 7)$, i.e., $24 < 29$, which is true. So $S(n)$ is true for $n = n_0 = 6$.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 6$, i.e., $4k < (k^2 - 7)$ for $k \geq 6$. Adding 4 on both sides,

$$\begin{aligned}4(k+1) = 4k + 4 &< (k^2 - 7) + 4 \\ &< (k^2 - 7) + (2k + 1),\end{aligned}$$

because when $k \geq 6$ we have $2k \geq 12 \Rightarrow 2k + 1 \geq 13 > 4$. Therefore

$$4(k+1) < (k^2 + 2k + 1) - 7, \quad \text{i.e.,} \quad 4(k+1) < (k+1)^2 - 7.$$

This is the statement $S(k+1)$. $\Rightarrow S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 6$. Thus $S(n)$ is true $\forall n \geq 6$. ■

Problem 6. Show that $2^n > n^2$ for all positive integers n greater than 4.

Solution. We prove $S(n)$: $2^n > n^2$ is true $\forall n \geq 5$.

Basic step: For $n = 5$: $S(5)$: $2^5 > 5^2$ (i.e., $32 > 25$), which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 5$, i.e., $2^k > k^2$. Multiplying by 2 on both sides,

$$2 \cdot 2^k = 2^{k+1} > 2k^2.$$

Since $k > 4$, we have

$$\begin{aligned} 2k^2 = k^2 + k^2 = k^2 + (k \times k) &> k^2 + 4k \\ &= k^2 + 2k + 2k \\ &> k^2 + 2k + 1 = (k+1)^2. \end{aligned}$$

Thus $2^{k+1} > (k+1)^2$ for $k > 4$. This is precisely the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 5$. Hence, by mathematical induction, $S(n)$ is true $\forall n > 4$. ■

Problem 7. By mathematical induction, prove that $n! \geq 2^{n-1}\ \forall n \geq 1$.

Solution. Let $S(n)$ be the given statement.

Basic step: For $n = 1$: $S(1):\ 1! \geq 2^{1-1}$, i.e., $1 \geq 1$, which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., $k! \geq 2^{k-1}$, i.e., $2^{k-1} \leq k!$. Multiplying by 2 on both sides,

$$\begin{aligned} 2^k = 2 \cdot 2^{k-1} &\leq 2 \cdot k! \\ &\leq (k+1) \cdot k! \qquad \text{because } (k+1) \geq 2 \text{ for } k \geq 1 \\ &= (k+1)! \end{aligned}$$

i.e., $(k+1)! \geq 2^k$, which is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 8. For all $n \in \mathbb{Z}^+$, show that if $n \geq 14$, then n can be written as a sum of 3's and/or 8's.

Solution. To prove: $S(n)$: n can be written as a sum of 3's and/or 8's, is true for all integers $n \geq 14$.

Basic step: We have $14 = (3+3)+8$. $\Rightarrow S(14)$ is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 14$. Then

$$k = (3+3+3+\cdots)+(8+8+\cdots).$$

Suppose this representation of k has r number of 3's and s number of 8's.

Case 1: $s \geq 1$ (at least one 8 is used). Then

$$\begin{aligned} k+1 &= \{\underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s}\} + 1 \\ &= \underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s-1} + 8 + 1 \\ &= \underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s-1} + (3+3+3) \\ &= \underbrace{(3+3+\cdots)}_{r+3} + \underbrace{(8+8+\cdots)}_{s-1}. \end{aligned}$$

Case 2: $s = 0$ (only 3's are used). Then $k = 3r \geq 14$, so $r \geq 5$. Replacing five 3's (sum 15) by two 8's (sum 16),

$$k+1 = \underbrace{(3+3+\cdots)}_{r-5} + (8+8).$$

$\Rightarrow (k+1)$ is a sum of 3's and/or 8's, i.e., $S(k+1)$ is true. Hence, by mathematical induction, $S(n)$ is true for all positive integers $n \geq 14$. ■

Problem 9. By mathematical induction, prove that for every positive integer n, the number $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8.

Solution. Let $S(n)$ be the given statement.

Basic step: $S(1):\ 5^1 + 2 \cdot 3^0 + 1 = 8$ is a multiple of 8. So, for $n = 1$, $S(n)$ is true.

Induction step: Assume that A_n is a multiple of 8 for $n = k \geq 1$, i.e., $A_k = 5^k + 2 \cdot 3^{k-1} + 1$ is a multiple of 8. We find

$$\begin{aligned} A_{k+1} - A_k &= (5^{k+1} + 2 \cdot 3^k + 1) - (5^k + 2 \cdot 3^{k-1} + 1) \\ &= (5^k \cdot 5 - 5^k) + 2(3^k - 3^{k-1}) \\ &= 5^k(5-1) + 2 \cdot 3^{k-1}(3-1) \\ &= 4 \cdot 5^k + 4 \cdot 3^{k-1} \\ &= 4(5^k + 3^{k-1}). \end{aligned}$$

5 and 3 are odd $\Rightarrow 5^k$ and 3^{k-1} are also odd $\Rightarrow 5^k + 3^{k-1}$ is even $\Rightarrow 4(5^k + 3^{k-1})$ is a multiple of 8, i.e., $A_{k+1} - A_k$ is a multiple of 8. Since A_k is a multiple of 8 (by assumption), A_{k+1} is also a multiple of 8. Thus, if A_n is a multiple of 8 for $n = k$, then A_n is a multiple of 8 for $n = k+1$. Hence, by mathematical induction, for every positive integer n, $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8. ■

Problem 10. QP Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n.

Solution. Let $A_n = 6^{n+2} + 7^{2n+1}$.

Basic step: $A_1 = 6^3 + 7^3 = 216 + 343 = 559 = 43 \times 13$ is divisible by 43. So A_n is divisible by 43 for $n = 1$.

Induction step: Assume that A_n is divisible by 43 for $n = k \geq 1$, i.e., $A_k = 6^{k+2} + 7^{2k+1}$ is divisible by 43. Then

$$\begin{aligned} A_{k+1} &= 6^{k+3} + 7^{2(k+1)+1} = 6^{k+3} + 7^{2k+3} \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 7^2 \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 49 \\ &= 6^{k+2} \cdot 6 + 7^{2k+1}(6 + 43) \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 6 + 7^{2k+1} \cdot 43 \qquad [a(b+c) = ab + ac] \\ &= 6[6^{k+2} + 7^{2k+1}] + 43 \cdot 7^{2k+1} \\ &= 6A_k + 43 \cdot 7^{2k+1}. \end{aligned}$$

By assumption, A_k is divisible by 43 $\Rightarrow 6A_k$ is also divisible by 43; and $43 \cdot 7^{2k+1}$ is divisible by 43. Therefore $6A_k + 43 \cdot 7^{2k+1}$ is divisible by 43 $\Rightarrow A_{k+1}$ is divisible by 43. Thus if $S(k)$ is true, then $S(k+1)$ is true, for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 11. QP Prove that $\displaystyle\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$ $\forall n \in \mathbb{Z}^+$, i.e.,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

Solution. Let $S(n)$ be the given statement.

Basic step: $S(1):\ \dfrac{1}{1(2)} = \dfrac{1}{1+1} \Rightarrow \dfrac{1}{2} = \dfrac{1}{2}$, which is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e.,

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1}.$$

Adding $\dfrac{1}{(k+1)(k+2)}$ on both sides,

$$\begin{aligned}\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)(k+2)} &= \frac{k}{k+1}+\frac{1}{(k+1)(k+2)} = \frac{1}{k+1}\left[k+\frac{1}{k+2}\right] \\ &= \frac{1}{k+1}\left[\frac{k^2+2k+1}{k+2}\right] = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2},\end{aligned}$$

which is precisely the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 12. QP Establish the following by mathematical induction: $\sum_{i=1}^{n} i2^i = 2+(n-1)2^{n+1}$.

Solution. Let $S(n)$ be the given statement.

Basic step: $S(1):\ 1\times 2^1 = 2+(1-1)2^2$, i.e., $2=2$, which is true.

Induction step: Assume that the result is true for $n=k$, where $k \geq 1$, i.e.,

$$S(k):\ 1\times 2^1+2\times 2^2+3\times 2^3+\cdots+k\times 2^k = 2+(k-1)2^{k+1}.$$

Adding $(k+1)\times 2^{k+1}$ on both sides,

$$\begin{aligned}1\times 2^1+2\times 2^2+\cdots+k\times 2^k+(k+1)\times 2^{k+1} &= 2+(k-1)2^{k+1}+(k+1)2^{k+1} \\ &= 2+2^{k+1}[k-1+k+1] \\ &= 2+2^{k+1}\cdot 2k \\ &= 2+k2^{k+2},\end{aligned}$$

which is precisely the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 1$. Hence $S(n)$ is true $\forall n \geq 1$. ■

Homework

Prove by mathematical induction:

1. $1^2+2^2+3^2+\cdots+n^2 = \frac{1}{6}n(n+1)(2n+1)$.
2. $1^3+2^3+3^3+\cdots+n^3 = \dfrac{n^2(n+1)^2}{4}$.
3. $2^n < n!$ for all positive integers $n > 3$. QP
4. For every integer n, 3 divides n^3-n.
5. For all $n \in \mathbb{Z}^+$, show that if $n \geq 24$, then n can be written as a sum of 5's and/or 7's. QP

Problem 13. QP Let $H_1 = 1,\ H_2 = 1+\frac{1}{2},\ H_3 = 1+\frac{1}{2}+\frac{1}{3},\ \ldots,\ H_n = 1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}$. Prove that

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n, \qquad \forall n \geq 1.$$

Solution. Let $S(n):\ \sum_{i=1}^{n} H_i = (n+1)H_n - n$, i.e., $H_1+H_2+H_3+\cdots+H_n = (n+1)H_n - n$.

Basic step: $S(1):\ H_1 = (1+1)H_1 - 1$, i.e., $1 = 2(1) - 1$, i.e., $1 = 1$. $\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e.,

$$S(k):\ H_1 + H_2 + \cdots + H_k = (k+1)H_k - k.$$

Adding H_{k+1} on both sides,

$$\begin{aligned} H_1 + H_2 + \cdots + H_k + H_{k+1} &= (k+1)H_k - k + H_{k+1} \\ &= (k+1)\left\{H_{k+1} - \frac{1}{k+1}\right\} - k + H_{k+1} \\ &\quad \left[\because\ H_{k+1} = H_k + \tfrac{1}{k+1},\ \text{so } H_k = H_{k+1} - \tfrac{1}{k+1}\ \text{(from the definition of } H_n)\right] \\ &= (k+1)H_{k+1} - 1 - k + H_{k+1} \\ &= (k+1+1)H_{k+1} - (k+1) \\ &= (k+2)H_{k+1} - (k+1). \end{aligned}$$

This is precisely the statement $S(k+1)$. By mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

2 Recursive Definitions

Consider $a_1, a_2, \ldots, a_n, \ldots$, a sequence in which a_1 is the first term, a_2 is the 2nd term, $\ldots$, a_n is the nth term. The element a_n is called the *general term* of the sequence, and the sequence is denoted by $\{a_n\}$. If the first term is denoted by a_0, then the nth term is denoted by a_{n-1} and the sequence is denoted by $\{a_{n-1}\}$.

For describing a sequence, two methods are commonly used:

(i) **Explicit method**

(ii) **Recursive method**

In the **explicit method**, the general term of the sequence is explicitly indicated.
Example: $E = \{a_n\}$, where $a_n = 2n$, $n \in \mathbb{Z}^+$.

In the **recursive method**, the first few terms of the sequence are explicitly indicated and the general term is specified through a rule (formula) which indicates how to obtain new terms of the sequence from the terms already known.
Example: Fibonacci numbers: $F_0 = 0,\ F_1 = 1,\ F_n = F_{n-1} + F_{n-2}\ \ \forall n \geq 2$.

> **Note:**
> 1. A recursive definition of a sequence should contain two parts: in the 1st part, a few terms of the sequence are indicated explicitly, and in the 2nd part, the rule to obtain new terms of the sequence is indicated.
> 2. **Recursive rule/formula:** a formula for a_n in terms of $a_{n-1}, a_{n-2}, \ldots$ (or) a formula for a_{n+1} in terms of $a_n, a_{n-1}, \ldots$

2.1 Problems

Problem 1. Obtain a recursive definition for the sequence $\{a_n\}$ in each of the following cases:

(i) $a_n = 6^n$

(ii) $a_n = n(n+3)$

(iii) $a_n = (n+1)!$

(iv) $a_n = 2 - (-1)^n$ QP

Solution. **(i)** $a_n = 6^n$.

$$\begin{aligned} a_1 &= 6^1 = 6 \\ a_2 &= 6^2 = 36 = 6 \times 6 = 6 \times a_1 \\ a_3 &= 6^3 = 216 = 6 \times 36 = 6 \times a_2 \\ a_4 &= 6^4 = 1296 = 6 \times 216 = 6 \times a_3 \\ &\vdots \end{aligned}$$

$a_1 = 6$; $\boxed{a_n = 6a_{n-1}}$ $\forall n \geq 2$ is the recursive definition (same as $a_{n+1} = 6a_n$ $\forall n \geq 1$).

(ii) $a_n = n(n+3)$.

$$\begin{aligned} a_1 &= 1(1+3) = 1 \times 4 = 4 \\ a_2 &= 2(2+3) = 2 \times 5 = 10 = a_1 + 6 = a_1 + (2 \times 3) \\ a_3 &= 3(3+3) = 3 \times 6 = 18 = a_2 + 8 = a_2 + (2 \times 4) \\ a_4 &= 4(4+3) = 4 \times 7 = 28 = a_3 + 10 = a_3 + (2 \times 5) \\ &\vdots \\ a_n &= a_{n-1} + 2(n+1) \quad \forall n \geq 2. \end{aligned}$$

Thus $a_1 = 4$, $a_n = a_{n-1} + 2(n+1)$ $\forall n \geq 2$ is the recursive definition.

(iii) $a_n = (n+1)!$.

$$\begin{aligned} a_1 &= (1+1)! = 2! = 2 & a_3 &= (3+1)! = 4! = 24 = a_2 \times 4 \\ a_2 &= (2+1)! = 3! = 6 = a_1 \times 3 & a_4 &= (4+1)! = 5! = 120 = a_3 \times 5 \end{aligned}$$

$\Rightarrow a_n = a_{n-1} \times (n+1)$ $\forall n \geq 2$. Thus $a_1 = 2$, $a_n = (n+1)\,a_{n-1}$ $\forall n \geq 2$ is the recursive definition.

(iv) $a_n = 2 - (-1)^n$.

$$\begin{aligned} a_1 &= 2 - (-1)^1 = 2 + 1 = 3 \\ a_2 &= 2 - (-1)^2 = 2 - 1 = 1 = 3 - 2 = a_1 - (-1)^2 \times 2 \\ a_3 &= 2 - (-1)^3 = 2 + 1 = 3 = 1 + 2 = a_2 - (-1)^3 \times 2 \\ a_4 &= 2 - (-1)^4 = 2 - 1 = 1 = 3 - 2 = a_3 - (-1)^4 \times 2 \\ &\vdots \end{aligned}$$

$a_1 = 3$, $a_n = a_{n-1} - 2(-1)^n$ $\forall n \geq 2$ is the recursive definition.

Problem 2. QP Find an explicit definition of the sequence defined recursively by $a_1 = 7$, $a_n = 2a_{n-1} + 1$ for $n \geq 2$.

Solution.

$$\begin{aligned} a_n &= 2a_{n-1} + 1 & &= 2^1 a_{n-1} + (2^1 - 1) \\ &= 2\{2a_{n-2} + 1\} + 1 = 4a_{n-2} + 3 & &= 2^2 a_{n-2} + (2^2 - 1) \\ &= 4\{2a_{n-3} + 1\} + 3 = 8a_{n-3} + 7 & &= 2^3 a_{n-3} + (2^3 - 1) \\ &= 8\{2a_{n-4} + 1\} + 7 = 16a_{n-4} + 15 & &= 2^4 a_{n-4} + (2^4 - 1) \\ &\vdots \end{aligned}$$

(Observe: the power of 2 and the subscript add up to n; when the subscript becomes 1, the power is $n-1$.)

$$\begin{aligned} \therefore\ a_n &= 2^{n-1} \cdot a_1 + 2^{n-1} - 1 = 2^{n-1}\{a_1 + 1\} - 1 \\ &= 2^{n-1}\{7+1\} - 1 = 2^{n-1}(8) - 1 = 2^{n-1} \cdot 2^3 - 1 = 2^{n+2} - 1 \\ &= \boxed{4 \times 2^n - 1}. \end{aligned}$$

Thus $a_n = 4 \times 2^n - 1$ is an explicit definition for the given sequence.

Problem 3. QP A sequence $\{a_n\}$ is defined recursively by $a_1 = 4,\ a_n = a_{n-1} + n\ \forall n \geq 2$. Find a_n in explicit form.

Solution.

$$\begin{aligned} a_n &= a_{n-1} + n \\ &= \{a_{n-2} + (n-1)\} + n \\ &= a_{n-3} + (n-2) + (n-1) + n \\ &= a_{n-4} + (n-3) + (n-2) + (n-1) + n \\ &\vdots \\ &= a_1 + 2 + 3 + 4 + \cdots + n \\ &= (a_1 - 1) + (1 + 2 + 3 + \cdots + n) \\ &= (a_1 - 1) + \frac{n(n+1)}{2}. \end{aligned}$$

$\boxed{a_n = 3 + \frac{n(n+1)}{2}}$ is the explicit formula for a_n.

Problem 4. QP A sequence $\{a_n\}$ is defined recursively by $a_0 = 1,\ a_1 = 1,\ a_2 = 1$ and $a_n = a_{n-1} + a_{n-3}$ $\forall n \geq 3$. Prove that $a_{n+2} \geq (\sqrt{2})^n\ \forall n \geq 0$.

Solution. Let $S(n):\ a_{n+2} \geq (\sqrt{2})^n$.

Basic step: First, $a_3 = a_2 + a_0 = 1 + 1 = 2$ and $a_4 = a_3 + a_1 = 2 + 1 = 3$.

$$\begin{aligned} S(0):&\ a_2 \geq (\sqrt{2})^0,\ \text{ i.e., } 1 \geq 1 \text{ — true} \\ S(1):&\ a_3 \geq (\sqrt{2})^1,\ \text{ i.e., } 2 \geq \sqrt{2} \text{ — true} \\ S(2):&\ a_4 \geq (\sqrt{2})^2,\ \text{ i.e., } 3 \geq 2 \text{ — true} \end{aligned}$$

Thus $S(n)$ is true for $n = 0, 1, 2$.

Induction step: Assume (alternate form of induction) that $S(0), S(1), \ldots, S(k)$ are true for some $k \geq 2$; in particular

$$a_{k+2} \geq (\sqrt{2})^k \quad \text{and} \quad a_k \geq (\sqrt{2})^{k-2}. \tag{1}$$

To prove: $a_{k+3} \geq (\sqrt{2})^{k+1}$. Consider

$$\begin{aligned} a_{k+3} &= a_{k+2} + a_k && \text{(using the definition of } a_n\text{)} \\ &\geq (\sqrt{2})^k + (\sqrt{2})^{k-2} && \text{(using (1))} \\ &= (\sqrt{2})^{k-2}\,[(\sqrt{2})^2 + 1] = (\sqrt{2})^{k-2}\,[3] \\ &= (\sqrt{2})^{k+1} \cdot (\sqrt{2})^{-3} \cdot 3 = (\sqrt{2})^{k+1} \cdot \frac{3}{2\sqrt{2}} \\ &\geq (\sqrt{2})^{k+1} && \left(\because\ \frac{3}{2\sqrt{2}} \geq 1\right). \end{aligned}$$

Thus $a_{k+3} \geq (\sqrt{2})^{k+1}$. Hence, by mathematical induction, the required result is true $\forall n \geq 0$. ■

Problem 5. QP For the sequence $\{a_n\}$ defined recursively by $a_1 = 8,\ a_2 = 22,\ a_n = 4(a_{n-1} - a_{n-2})$ for $n \geq 3$, prove that $a_n = (5 + 3n)\,2^{n-1}$ for $n \geq 1$.

Solution. $S(n):\ a_n = (5 + 3n)\,2^{n-1}\ \forall n \geq 1$.

Basic step: $S(1):\ a_1 = (5+3)\,2^0 = 8.$ $S(2):\ a_2 = (5+6)\,2 = 22.$ So $S(n)$ is true for $n = 1, 2.$

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$, i.e.,

$$a_k = (5+3k)\,2^{k-1}, \qquad a_{k-1} = \{5+3(k-1)\}\,2^{k-2}. \tag{1}$$

To prove: $a_{k+1} = \{5+3(k+1)\}\,2^{k+1-1}$, i.e., $a_{k+1} = (8+3k)\,2^k$. Consider

$$\begin{aligned}
a_{k+1} &= 4(a_k - a_{k-1}) \qquad \text{(using the definition of } a_n\text{)} \\
&= 4\left[(5+3k)\,2^{k-1} - \{5+3(k-1)\}\,2^{k-2}\right] \qquad \text{(using (1))} \\
&= 4\left[5\,(2^{k-1} - 2^{k-2}) + 3\{k\cdot 2^{k-1} - (k-1)\,2^{k-2}\}\right] \\
&= 4\left[5\cdot 2^{k-2}(2-1) + 3\cdot 2^{k-2}\{2k-k+1\}\right] \\
&= 4\left[5\cdot 2^{k-2} + 3\cdot 2^{k-2}(k+1)\right] \\
&= 4\cdot 2^{k-2}\,[5+3k+3] = 2^2\cdot 2^{k-2}\,[8+3k] \\
&= 2^k\,(8+3k),
\end{aligned}$$

which is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

Problem 6. QP A sequence $\{c_n\}$ is defined recursively by $c_n = 3c_{n-1} - 2c_{n-2}\ \forall n \geq 3$, with $c_1 = 5$ and $c_2 = 3$. Show that $c_n = -2^n + 7$.

Solution. Let $S(n):\ c_n = -2^n + 7\ \forall n \geq 1.$

Basic step: $S(1):\ c_1 = -2^1 + 7$, i.e., $5 = 5$ — true. $S(2):\ c_2 = -2^2 + 7$, i.e., $3 = 3$ — true.
$\therefore S(n)$ is true for $n = 1, 2.$

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$: $c_k = -2^k + 7$ and $c_{k-1} = -2^{k-1} + 7$.
To prove: $S(k+1):\ c_{k+1} = -2^{k+1} + 7$. Consider

$$\begin{aligned}
c_{k+1} &= 3c_k - 2c_{k-1} \qquad \text{(by the definition of } c_n\text{)} \\
&= 3(-2^k + 7) - 2(-2^{k-1} + 7) \\
&= (-3\cdot 2^k + 2\cdot 2^{k-1}) + 21 - 14 \\
&= 2^{k-1}(-3\cdot 2 + 2) + 7 \\
&= -4\cdot 2^{k-1} + 7 = -2^2\cdot 2^{k-1} + 7 \\
&= -2^{k+1} + 7,
\end{aligned}$$

which is exactly the statement $S(k+1)$. Thus, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

Problem 7. QP The Fibonacci numbers are defined by $F_0 = 0,\ F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}\ \forall n \geq 2$. Prove that

$$\sum_{i=0}^{n} F_i^2 = F_n \times F_{n+1}.$$

Solution. Let $S(n):\ \sum_{i=0}^{n} F_i^2 = F_n \times F_{n+1}.$

Basic step: $S(0):\ F_0^2 = F_0 \times F_1$, i.e., $0^2 = 0 \times 1$, i.e., $0 = 0.$
$S(1):\ F_0^2 + F_1^2 = F_1 \times F_2$, i.e., $0 + 1^2 = 1 \times (F_1 + F_0) = 1 \times (1+0)$, i.e., $1 = 1.$
$S(n)$ is true for $n = 0, 1.$

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e.,

$$S(k):\ \sum_{i=0}^{k} F_i^2 = F_k \times F_{k+1}. \tag{1}$$

To prove: $S(k+1):\ \sum_{i=0}^{k+1} F_i^2 = F_{k+1} \times F_{k+2}$. Consider

$$\begin{aligned}\sum_{i=0}^{k+1} F_i^2 = F_0^2 + F_1^2 + \cdots + F_k^2 + F_{k+1}^2 &= \sum_{i=0}^{k} F_i^2 + F_{k+1}^2 \\ &= (F_k \times F_{k+1}) + F_{k+1}^2 && \text{(using (1))} \\ &= F_{k+1}(F_k + F_{k+1}) \\ &= F_{k+1}(F_{k+2}) && (\because F_n = F_{n-1} + F_{n-2}) \\ &= F_{k+1} \times F_{k+2}.\end{aligned}$$

This is exactly the statement $S(k+1)$. Thus, by mathematical induction, $S(n)$ is true $\forall n \geq 0$. ■

Problem 8. QP For the Fibonacci sequence $F_0, F_1, F_2, \ldots$ prove that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

Solution. Let $S(n)$ denote the given statement.

Basic step: $S(0):\ F_0 = \frac{1}{\sqrt{5}}[1-1] = 0.$

$$S(1):\ F_1 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right) - \left(\frac{1-\sqrt{5}}{2}\right)\right] = \frac{1}{\sqrt{5}}\left[\frac{1}{2} + \frac{\sqrt{5}}{2} - \frac{1}{2} + \frac{\sqrt{5}}{2}\right] = \frac{1}{\sqrt{5}} \cdot \frac{2\sqrt{5}}{2} = 1.$$

Thus $S(n)$ is true for $n = 0, 1$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 1$, i.e.,

$$F_k = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right] \text{ (and similarly for } F_{k-1}).$$

To prove: $S(k+1)$ is true, i.e., $F_{k+1} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right]$.

Consider $F_{k+1} = F_k + F_{k-1}$ (by the definition of Fibonacci numbers).

$$\begin{aligned}\Rightarrow F_{k+1} &= \frac{1}{\sqrt{5}}\left[\left(\tfrac{1+\sqrt{5}}{2}\right)^k - \left(\tfrac{1-\sqrt{5}}{2}\right)^k\right] + \frac{1}{\sqrt{5}}\left[\left(\tfrac{1+\sqrt{5}}{2}\right)^{k-1} - \left(\tfrac{1-\sqrt{5}}{2}\right)^{k-1}\right] \\ &= \frac{1}{\sqrt{5}}\left[\left(\tfrac{1+\sqrt{5}}{2}\right)^{k-1}\left\{\tfrac{1+\sqrt{5}}{2} + 1\right\} - \left(\tfrac{1-\sqrt{5}}{2}\right)^{k-1}\left\{\tfrac{1-\sqrt{5}}{2} + 1\right\}\right] \\ &= \frac{1}{\sqrt{5}}\left[\left(\tfrac{1+\sqrt{5}}{2}\right)^{k-1}\left(\tfrac{3+\sqrt{5}}{2}\right) - \left(\tfrac{1-\sqrt{5}}{2}\right)^{k-1}\left(\tfrac{3-\sqrt{5}}{2}\right)\right] \\ &= \frac{1}{\sqrt{5}}\left[\left(\tfrac{1+\sqrt{5}}{2}\right)^{k-1}\left(\tfrac{1+\sqrt{5}}{2}\right)^2 - \left(\tfrac{1-\sqrt{5}}{2}\right)^{k-1}\left(\tfrac{1-\sqrt{5}}{2}\right)^2\right]\end{aligned}$$

$$\left[\because \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+5+2\sqrt{5}}{4} = \frac{6+2\sqrt{5}}{4} = \frac{2(3+\sqrt{5})}{4} = \frac{3+\sqrt{5}}{2};\ \text{similarly } \left(\frac{1-\sqrt{5}}{2}\right)^2 = \frac{3-\sqrt{5}}{2}\right]$$

$$\therefore F_{k+1} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right].$$

Thus $S(k+1)$ is true whenever $S(k-1)$ and $S(k)$ are true. By mathematical induction, the result is true $\forall n \geq 0$. ■

Problem 9. QP Let F_n denote the Fibonacci numbers. Prove that

$$\sum_{i=1}^{n} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{n+2}}{2^n}.$$

Solution. Let $S(n)$ denote the given statement.

Basic step: For $n = 1$: $S(1): \dfrac{F_0}{2^1} = 1 - \dfrac{F_3}{2^1}$.

We have $F_0 = 0,\ F_1 = 1,\ F_2 = F_1 + F_0 = 1,\ F_3 = F_2 + F_1 = 1 + 1 = 2$. $\therefore S(1): \dfrac{0}{2} = 1 - \dfrac{2}{2} \Rightarrow 0 = 0$. $\therefore S(n)$ is true for $n = 1$.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e.,

$$S(k): \sum_{i=1}^{k} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{k+2}}{2^k}. \tag{1}$$

To prove: $S(k+1): \sum_{i=1}^{k+1} \dfrac{F_{i-1}}{2^i} = 1 - \dfrac{F_{k+3}}{2^{k+1}}$. Consider

$$\sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = \underbrace{\frac{F_0}{2} + \frac{F_1}{2^2} + \frac{F_2}{2^3} + \cdots + \frac{F_{k-1}}{2^k}}_{\sum_{i=1}^{k} F_{i-1}/2^i} + \frac{F_k}{2^{k+1}}.$$

$$\Rightarrow \sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = \left(1 - \frac{F_{k+2}}{2^k}\right) + \frac{F_k}{2^{k+1}} \qquad \text{(using (1))}$$

$$= 1 - \frac{1}{2^k}\left[F_{k+2} - \frac{F_k}{2}\right] = 1 - \frac{1}{2^k}\left[\frac{2F_{k+2} - F_k}{2}\right]$$

$$= 1 - \frac{1}{2^{k+1}}\left[F_{k+2} + F_{k+2} - F_k\right].$$

But $F_{k+2} = F_{k+1} + F_k \Rightarrow F_{k+2} - F_k = F_{k+1}$.

$$\therefore \sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = 1 - \frac{1}{2^{k+1}}\left[F_{k+2} + F_{k+1}\right]$$

$$= 1 - \frac{1}{2^{k+1}} F_{k+3} \qquad (\because F_{k+3} = F_{k+2} + F_{k+1}).$$

This is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

Problem 10. QP If F_i's are Fibonacci numbers defined recursively by $F_0 = 0,\ F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, and L_i's are Lucas numbers defined recursively as $L_0 = 2,\ L_1 = 1$ and $L_n = L_{n-1} + L_{n-2}\ \forall n \geq 2$, then prove that

$$L_n = F_{n-1} + F_{n+1} \qquad \forall n \geq 1.$$

Solution. Let $S(n): L_n = F_{n-1} + F_{n+1}\ \forall n \geq 1$.

Basic step: $S(1): L_1 = F_0 + F_2$, i.e., $1 = 0 + 1 = 1$ $(\because F_2 = F_1 + F_0 = 1 + 0 = 1)$.
Also $S(2): L_2 = L_1 + L_0 = 3$ and $F_1 + F_3 = 1 + 2 = 3$. $\therefore S(n)$ is true for $n = 1, 2$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$:

$$L_k = F_{k-1} + F_{k+1}, \qquad L_{k-1} = F_{k-2} + F_k. \tag{1}$$

To prove: $S(k+1):\ L_{k+1} = F_k + F_{k+2}$. Consider

$$\begin{aligned} L_{k+1} &= L_k + L_{k-1} && \text{(by the definition of } L_n\text{)} \\ \Rightarrow L_{k+1} &= (F_{k-1} + F_{k+1}) + (F_{k-2} + F_k) && \text{(by the assumption made)} \\ &= (F_{k-1} + F_{k-2}) + (F_{k+1} + F_k) \\ &= F_k + F_{k+2} && \text{(by the definition of } F_n\text{).} \end{aligned}$$

This is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

Problem 11. The Ackermann's numbers are defined recursively for $m, n \in \mathbb{N}$ as follows:

$$A_{0,n} = n+1 \qquad \text{for } n \geq 0 \tag{1}$$
$$A_{m,0} = A_{m-1,1} \qquad \text{for } m > 0 \tag{2}$$
$$A_{m,n} = A_{m-1,p},\ \text{where } p = A_{m,n-1} \qquad \text{for } m, n > 0 \tag{3}$$

Prove that $A_{1,n} = n+2\ \forall n \in \mathbb{N}$.

Solution. Let $S(n):\ A_{1,n} = n+2\ \forall n \in \mathbb{N}$.

Basic step: $S(1):\ A_{1,1} = 1+2 \Rightarrow \text{RHS} = 3$.
Consider LHS: $A_{1,1} = A_{0,p}$, where $p = A_{1,0} = A_{0,1} = 1+1 = 2$ (by (2) and (1)).

$$\therefore A_{1,1} = A_{0,2} = 2+1 = 3 \quad \text{(by (1)).}$$

LHS $= 3 =$ RHS. $\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e.,

$$A_{1,k} = k+2. \tag{4}$$

To prove: $S(k+1):\ A_{1,k+1} = k+3$. Consider

$$\begin{aligned} A_{1,k+1} &= A_{0,p}\ \text{(using (3)), where } p = A_{1,k} = k+2\ \text{(using (4))} \\ &= A_{0,k+2} \\ &= k+2+1 \qquad \text{(using (1))} \\ &= k+3, \end{aligned}$$

which is the statement $S(k+1)$. Hence, by mathematical induction, $A_{1,n} = n+2\ \forall n \in \mathbb{N}$. ■

3 Fundamental Principles of Counting

3.1 The Rules of Sum and Product

The Sum Rule

If T_1 and T_2 are two tasks such that T_1 can be performed in m different ways, T_2 can be performed in n different ways, and these two tasks cannot be performed simultaneously, then one of the tasks (T_1 *or* T_2) can be performed in $m+n$ ways.

Example: Suppose a library has 12 books on Mathematics, 10 books on Physics and 16 books on Computers. If a student wishes to choose one of these books to study, then the number of ways he can choose the book is $12+10+16 = 38$.

The Product Rule

If T_1 and T_2 are two tasks such that T_1 can be performed in m different ways and, for each of these ways, T_2 can be performed in n different ways, then both the tasks can be performed in $m \times n$ different ways.

Example: Suppose a person has 8 shirts and 5 ties. Then the number of ways of choosing a shirt and a tie is $8 \times 5 = 40$.

Problems

Problem 1. A licence plate consists of two English letters followed by 4 digits. If repetitions are allowed, how many of the plates have only vowels and even digits?

Solution. Each of the first 2 positions can be filled in 5 ways (with the 5 vowels), and each of the remaining 4 places can be filled in 5 ways (with the digits $0,2,4,6,8$). $\therefore$ Number of possible licence plates of the given type is

$$(5 \times 5) \times (5 \times 5 \times 5 \times 5) = 5^6 = 15{,}625.$$

Problem 2. QP Cars of a particular company come in 4 models, 12 colours, 3 engine sizes and 2 transmission types. (a) How many distinct cars of this company can be manufactured? (b) Of these, how many have the same colour?

Solution. (a) The number of distinct cars that can be manufactured is $4 \times 12 \times 3 \times 2 = 288$.
(b) For any chosen colour, the number of distinct cars that can be manufactured is $4 \times 3 \times 2 = 24$.

Problem 3. Find the number of 3-digit even numbers with no repeated digits.

Solution. Consider the number in the form xyz, where each of x, y, z represents a digit. Since xyz has to be even, z has to be $0, 2, 4, 6$ or 8.

- If $z = 0$ (1 choice only), then x has 9 choices and y has 8 choices.
- If $z = 2, 4, 6$ or 8 (4 choices), then x has 8 choices and y has 8 choices.
  [Note that y can take 0, but x cannot take 0.]

$\therefore$ The desired number is $(9 \times 8 \times 1) + (8 \times 8 \times 4) = 72 + 256 = 328$.

Problem 4. QP A bit is either 0 or 1. A byte is a sequence of 8 bits. Find (i) the number of bytes; (ii) the number of bytes that begin with 11 and end with 11; (iii) the number of bytes that begin with 11 and do not end with 11; (iv) the number of bytes that begin with 11 or end with 11.

Solution. (i) Since each byte contains 8 bits and each bit is a 0 or 1 (two choices), the number of bytes is $2 \times 2 \times \cdots \times 2$ (8 times) $= 2^8 = 256$.

(ii) In a byte beginning and ending with 11 (11____11), there are 4 empty positions. These can be filled in $2^4 = 16$ ways. $\therefore$ There are 16 bytes which begin and end with 11.

(iii) In a byte beginning with 11, there are 6 empty positions. These can be filled in $2^6 = 64$ ways, so there are 64 bytes that begin with 11. Since 16 bytes begin and end with 11, the number of bytes that begin with 11 but do not end with 11 is $64 - 16 = 48$.

(iv) The number of bytes that end with 11 is also 64 (as in (iii)), and the number of bytes that begin and end with 11 is 16. $\therefore$ The number of bytes that begin with 11 or end with 11 is

$$64 + 64 - 16 = 112 \qquad [\because \text{ for any two finite sets } A, B: \ |A \cup B| = |A| + |B| - |A \cap B|].$$

> Let A: bytes that begin with 11, B: bytes that end with 11. Then $A \cap B$: begin and end with 11; $A \cup B$: begin or end with 11.

(iii) asks for $|A \cap \overline{B}| = |A| - |A \cap B|$; (iv) asks for $|A \cup B| = |A| + |B| - |A \cap B|$.

Problem 5. Find the number of proper divisors of 441000. (Proper divisors are all divisors excluding 1 and the number itself.)

Solution. We have $441000 = 2^3 \cdot 3^2 \cdot 5^3 \cdot 7^2$.

$\therefore$ Every divisor of $n = 441000$ must be of the form $d = 2^p 3^q 5^r 7^s$, where $0 \le p \le 3,\ 0 \le q \le 2,\ 0 \le r \le 3,\ 0 \le s \le 2$.
So p can be chosen in 4 ways, q in 3 ways, r in 4 ways and s in 3 ways.
Thus the total number of divisors $= 4 \times 3 \times 4 \times 3 = 144$.
The total number of proper divisors $= 144 - 2 = 142$
[-2 because a number is always divisible by 1 and by itself; so we leave out these 2 choices].

2	441000
2	220500
2	110250
3	55125
3	18375
5	6125
5	1225
5	245
7	49
7	7
	1

Problem 6. Licence plates consist of 2 letters followed by 4 digits. In how many ways can this be done if (i) no letter or digit can be repeated; (ii) repetitions are allowed; (iii) only vowels and even digits can be used (with repetitions)?

Solution. (i) There are 26 letters in English: the 1st place has 26 choices and the 2nd place has 25 choices (since no repetition is allowed). Similarly, the digits are 10 in number $(0, 1, \ldots, 9)$: the 1st place can be filled in 10 ways, the 2nd in 9, the 3rd in 8 and the 4th in 7 ways. $\therefore$ Total number of ways $= 26 \times 25 \times 10 \times 9 \times 8 \times 7 = 32{,}76{,}000$.

(ii) Since repetitions are allowed: $(26 \times 26) \times (10 \times 10 \times 10 \times 10) = 26^2 \times 10^4 = 67{,}60{,}000$.

(iii) There are 5 vowels to be used, and the even digits are $0, 2, 4, 6, 8$ (5 choices). $\therefore$ $(5 \times 5) \times (5 \times 5 \times 5 \times 5) = 5^6 = 15{,}625$.

Problem 7. Find the total number of positive integers that can be formed from the digits $1, 2, 3, 4$ if no digit is repeated in any one integer.

Solution. Let s_1, s_2, s_3, s_4 denote the number of integers containing 1 digit, 2 digits, 3 digits and 4 digits respectively.

- There are 4 integers containing exactly 1 digit (1, 2, 3 or 4): $\boxed{s_1 = 4}$.
- A 2-digit integer without repetition can be chosen in $4 \times 3 = 12$ ways: $\boxed{s_2 = 12}$.
- A 3-digit integer without repetition can be chosen in $4 \times 3 \times 2 = 24$ ways: $\boxed{s_3 = 24}$.
- Similarly, $s_4 = 4 \times 3 \times 2 \times 1$: $\boxed{s_4 = 24}$.

Thus the required number is $s_1 + s_2 + s_3 + s_4 = 4 + 12 + 24 + 24 = 64$.

Problem 8. QP Determine the number of 6-digit integers (no leading 0's) in which (i) no digit is repeated; (ii) no digit is repeated and it is even; (iii) no digit is repeated and it is divisible by 5.

Solution. (i) The first place cannot be zero (9 choices); the second place can be zero (9 choices); and so on:

$$9 \times 9 \times 8 \times 7 \times 6 \times 5 = 1{,}36{,}080.$$

(ii) There are 2 possibilities for the unit (last) place.
Case 1: If the last place is 0 (1 choice): $9 \times 8 \times 7 \times 6 \times 5 \times 1 = 15{,}120 = s_1$ (say).

Case 2: If the last place is not 0, i.e., the unit place takes $2,4,6,8$ (4 choices), then the first place has 8 choices (no zero, and not the unit digit):

$$8\times 8\times 7\times 6\times 5\times 4 = 53{,}760 = s_2 \text{ (say)}.$$

Thus the required number $= s_1 + s_2 = 15{,}120 + 53{,}760 = 68{,}880$.

(iii) *Case 1:* If the unit place is 0: $9\times 8\times 7\times 6\times 5\times 1 = 15{,}120 = s_3$.
Case 2: If the unit place is not zero, i.e., it is 5 (1 choice); the first place cannot be 0 or 5 (8 choices):

$$8\times 8\times 7\times 6\times 5\times 1 = 13{,}440 = s_4.$$

$\therefore$ Required number $= s_3 + s_4 = 15{,}120 + 13{,}440 = 28{,}560$.

3.2 Permutations

A linear arrangement of distinct objects is called a *permutation*. The number of arrangements of r distinct objects out of n objects is denoted by $P(n,r)$ or ${}^nP_r$, and

$$P(n,r) = n(n-1)(n-2)\cdots\big(n-(r-1)\big) = \frac{n!}{(n-r)!}.$$

Note:

1. The number of different arrangements of n objects taken all n at a time is $n!$. This is the number of permutations of n distinct objects.
2. Suppose we wish to find the number of permutations that can be formed from a collection of n objects of which n_1 are of one type, n_2 are of a 2nd type, n_3 of a 3rd type, $\ldots$, n_k are of the kth type, with $n_1 + n_2 + \cdots + n_k = n$. Then the number of permutations of the n objects taken all at a time is
$$\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$
3. The number of circular arrangements of n different objects is $(n-1)!$.

Problems

Problem 1. 5 men and 4 women sit in a row such that the women occupy the even places. How many such arrangements are possible?

Solution. The 5 men can be seated (in the odd places) in 5! ways and the 4 women can be placed in the even places in 4! ways. $\therefore$ Total number of arrangements $= 5!\times 4! = 120\times 24 = 2880$.

Problem 2. Find n if (i) $P(n,2) = 90$; (ii) $P(n,3) = 3P(n,2)$.

Solution. (i) $P(n,2) = n(n-1) = 90$

$$\Rightarrow n^2 - n - 90 = 0 \Rightarrow n^2 - 10n + 9n - 90 = 0$$
$$\Rightarrow n(n-10) + 9(n-10) = 0 \Rightarrow (n-10)(n+9) = 0$$
$$\Rightarrow n = 10 \text{ or } -9. \qquad \therefore \boxed{n = 10} \ (n \text{ must be positive}).$$

(ii) $P(n,3) = 3P(n,2)$

$$\Rightarrow \frac{n!}{(n-3)!} = 3\cdot\frac{n!}{(n-2)!} \Rightarrow \frac{(n-2)!}{(n-3)!} = 3 \Rightarrow \frac{(n-2)(n-3)!}{(n-3)!} = 3 \Rightarrow n-2 = 3 \Rightarrow \boxed{n = 5}.$$

Problem 3. How many different strings (sequences) of length 4 can be formed using the letters of the word FLOWER?

Solution. FLOWER has 6 letters, all of which are distinct. The required number of strings is the number of permutations of these six letters chosen 4 at a time:

$$P(6,4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = \frac{6\times5\times4\times3\times2!}{2!} = 360.$$

Problem 4. Find the number of distinguishable permutations of the letters in the following words: (i) BASIC (ii) CALCULUS (iii) MATHEMATICS (iv) STRUCTURES (v) ENGINEERING

Solution. (i) The word BASIC has 5 letters $(n = 5)$, all letters being distinct. $\therefore$ Number of permutations of BASIC $= 5! = 120$.

(ii) The word CALCULUS has 8 letters $(n = 8)$, of which 2 are C, 2 are L, 2 are U, and 1 each are A and S.

$$\therefore \text{ Required number of permutations} = \frac{8!}{2!2!2!1!1!} = 5040.$$

(iii) The word MATHEMATICS has 11 letters $(n = 11)$, in which M $= 2$, A $= 2$, T $= 2$, H $= 1$, E $= 1$, I $= 1$, C $= 1$, S $= 1$.

$$\therefore \text{ Number of permutations} = \frac{11!}{2!\times2!\times2!\times(1!)^5} = 49{,}89{,}600.$$

(iv) STRUCTURES has 10 letters $(n = 10)$, where S $= 2$, T $= 2$, R $= 2$, U $= 2$, C $= 1$, E $= 1$.

$$\therefore \text{ Number of permutations} = \frac{10!}{2!2!2!2!1!1!} = 2{,}26{,}800.$$

(v) ENGINEERING has 11 letters $(n = 11)$, where E $= 3$, N $= 3$, G $= 2$, I $= 2$, R $= 1$.

$$\therefore \text{ Number of permutations} = \frac{11!}{3!3!2!2!1!} = 2{,}77{,}200.$$

Problem 5. Find the number of permutations of the letters of the word MASSASAUGA. In how many of these are all four A's together? How many of them begin with S?

Solution. (i) The given word has 10 letters, of which 4 are A, 3 are S and 1 each are M, U, G.

$$\therefore \text{ Required number of permutations} = \frac{10!}{4!3!1!1!1!} = 25{,}200.$$

(ii) If, in a permutation, all A's are to be together, we treat all the A's as one single letter, say $X = (\text{AAAA})$. Then the letters to be permuted read X, S, S, S, M, U, G, which are 7 in number.

$$\therefore \text{ Number of permutations} = \frac{7!}{1!3!1!1!1!} = 840.$$

(iii) For permutations beginning with S, there occur nine empty positions to fill, where two are S, four are A and one each are M, U, G.

$$\therefore \text{ Number of such permutations} = \frac{9!}{2!4!(1!)^3} = 7560.$$

Problem 6. Find the number of permutations of the letters of the word MISSISSIPPI. How many of these begin with an I? How many of these begin and end with an S?

Solution. (i) The given word has 11 letters, in which there are 4 I's, 4 S's, 2 P's and 1 M.

$$\therefore \text{ Number of permutations} = \frac{11!}{4!4!2!1!} = 34{,}650.$$

(ii) For permutations beginning with I, there occur 10 empty positions to fill, where 3 are I, 4 are S, 2 are P and 1 is M.

$$\therefore \text{ Number of such permutations} = \frac{10!}{3!4!2!1!} = 12{,}600.$$

(iii) For permutations beginning and ending with S, there occur 9 empty positions to fill, where 2 are S, 4 are I, 2 are P and 1 is M.

$$\therefore \text{ Number of such permutations} = \frac{9!}{2!4!2!1!} = 3780.$$

Problem 7. (a) How many arrangements are there for all letters in the word SOCIOLOGICAL? (b) In how many of these arrangements (i) are A and G adjacent? (ii) are all the vowels adjacent?

Solution. (a) The given word has 12 letters, of which 3 are O; 2 each are C, I, L; and 1 each are S, A, G.

$$\therefore \text{ Number of arrangements} = \frac{12!}{3!2!2!2!(1!)^3} = 99{,}79{,}200.$$

(b)(i) If A and G are to be adjacent, we treat A and G together as a single letter, say X, so that there are 3 O's, 2 C's, 2 I's, 2 L's, 1 S and 1 X, totalling 11 letters. These can be arranged in

$$\frac{11!}{3!2!2!2!1!1!} = 8{,}31{,}600 \text{ ways.}$$

Also, the letters A and G can be arranged among themselves in $\frac{2!}{1!1!} = 2$ ways. $\therefore$ Total number of arrangements in this case $= 8{,}31{,}600 \times 2 = 16{,}63{,}200$.

(ii) If all the vowels are to be adjacent, we treat all the vowels present in the word (namely A, O, I) as a single letter, say Y, so that there are 2 C's, 2 L's, 1 S, 1 G and 1 Y, totalling 7 letters. These can be arranged in

$$\frac{7!}{2!2!1!1!1!} = 1260 \text{ ways.}$$

Also, since the word contains 3 O's, 2 I's and 1 A (totalling 6), the letters O, I, A (clubbed as Y) can be arranged among themselves in $\frac{6!}{3!2!1!} = 60$ ways. $\therefore$ Total number of arrangements in this case $= 1260 \times 60 = 75{,}600$.

Problem 8. How many six-digit numbers can one make using the digits $1,3,3,7,7,8$?

Solution. Here $n = 6$. Among the given digits there is one 1 (1st type), two 3's (2nd type), two 7's (3rd type) and one 8 (4th type).

$$\text{Number of six-digit numbers} = \frac{6!}{1!2!2!1!} = \frac{6\times5\times4\times3\times2}{4} = 30\times6 = 180.$$

Problem 9. How many positive integers n can we form using the digits $3,4,4,5,5,6,7$ if we want n to exceed 50,00,000?

Solution. Let $n = x_1x_2x_3x_4x_5x_6x_7$, where $x_1, x_2, \ldots, x_7$ are the given digits. Since n exceeds 50,00,000, x_1 can take 5, 6 or 7.

- If $x_1 = 5$, then $x_2 \ldots x_7$ is an arrangement of the remaining 6 digits, which contain two 4's and one each of 3, 5, 6, 7. Number of such arrangements $= \frac{6!}{2!1!1!1!1!} = 360$.

- If $x_1 = 6$, then $x_2 \ldots x_7$ is an arrangement of the remaining 6 digits, which contain two 4's, two 5's and one each of 3, 7. Number of such arrangements $= \dfrac{6!}{2!2!1!1!} = 180$.
- If $x_1 = 7$, then $x_2 \ldots x_7$ is an arrangement of the remaining 6 digits, which contain two 4's, two 5's and one each of 3, 6. Number of such arrangements $= \dfrac{6!}{2!2!1!1!} = 180$.

$\therefore$ By the sum rule, the number of n's of the desired type is $360 + 180 + 180 = 720$.

Problem 10. In how many ways can 6 men and 6 women be seated in a row (i) if any person may sit next to any other? (ii) if men and women must occupy alternate seats?

Solution. (i) If any person may sit next to any other, no distinction is made between men and women in their seating. $\therefore$ Number of ways they can be seated $= 12!\ [= (6+6)!] = 47{,}90{,}01{,}600$.

(ii) If men and women are to occupy alternate seats, the 6 men can be seated in the odd places in 6! ways and the 6 women in the even places in 6! ways. $\therefore$ Number of ways in which men occupy odd places and women occupy even places $= 6! \times 6! = 5{,}18{,}400$. Similarly, the number of ways in which men occupy even places and women occupy odd places $= 6! \times 6! = 5{,}18{,}400$. $\therefore$ Total number of ways men and women occupy alternate seats $= 5{,}18{,}400 + 5{,}18{,}400 = 10{,}36{,}800$.

Problem 11. For non-negative integers n and r, if $n+1 > r$, prove that

$$P(n+1, r) = \frac{(n+1)}{(n+1-r)} P(n, r).$$

Solution. We have $P(n,r) = \dfrac{n!}{(n-r)!}$ and $P(n+1,r) = \dfrac{(n+1)!}{(n+1-r)!}$.

$$\therefore \frac{P(n+1,r)}{P(n,r)} = \frac{(n+1)!}{(n+1-r)!} \times \frac{(n-r)!}{n!} = \frac{(n+1)!}{n!} \times \frac{(n-r)!}{(n+1-r)!}$$
$$= \frac{(n+1)\cdot n!}{n!} \times \frac{(n-r)!}{(n+1-r)(n-r)!} = \frac{(n+1)}{(n+1-r)}.$$

Thus $P(n+1,r) = \left(\dfrac{n+1}{n+1-r}\right) P(n,r)$. ■

Problem 12. 4 different Mathematics books, 5 different CO books and 2 different GTC books are to be arranged in a shelf. How many different arrangements are possible if (a) the books in each particular subject must all be together? (b) only the Mathematics books must be together?

Solution. (a) The Mathematics books can be arranged among themselves in 4! ways, the CO books in 5! ways, the GTC books in 2! ways, and the 3 groups in 3! ways.

$$\therefore \text{ Number of possible arrangements} = 4! \times 5! \times 2! \times 3! = 34{,}560.$$

(b) Since the Mathematics books have to be together, consider the 4 Mathematics books as one single book. Then we have 8 books, which can be arranged in 8! ways. But the Mathematics books can be arranged among themselves in 4! ways.

$$\therefore \text{ Required number of arrangements} = 8! \times 4! = 9{,}67{,}680.$$

3.3 Combinations

Selecting a set of r objects from a set of n objects $(n \geq r)$ without regard to order is called a *combination* of r objects. The total number of combinations of r different objects that can be selected from n different objects is denoted by $C(n,r)$ or ${}^nC_r$ and is given by

$${}^nC_r = \frac{n!}{(n-r)!\,r!}.$$

Note: ${}^nC_r = {}^nC_{n-r}$.

Problems

Problem 1. How many committees of 5 members with one chairperson can be selected from 12 persons?

Solution. 1 chairperson out of 12 can be selected in 12 ways. The remaining 4 members out of 11 persons can be selected in ${}^{11}C_4$ ways. $\therefore$ Required number of committees $= 12 \times {}^{11}C_4 = 12 \times 330 = 3960$.

Problem 2. A certain question paper contains 3 parts A, B, C with 4 questions in Part A, 5 in Part B and 6 in Part C. It is required to answer 7 questions, selecting at least 2 from each part. In how many different ways can a student select his questions for answering?

Solution.

	No. of questions	No. of questions chosen		
Part	it contains	Case (1)	Case (2)	Case (3)
A	4	2	2	3
B	5	2	3	2
C	6	3	2	2
No. of ways of selection		${}^4C_2 \times {}^5C_2 \times {}^6C_3$ $(S_1 = 1200)$	${}^4C_2 \times {}^5C_3 \times {}^6C_2$ $(S_2 = 900)$	${}^4C_3 \times {}^5C_2 \times {}^6C_2$ $(S_3 = 600)$

$\therefore$ Total number of ways $= S_1 + S_2 + S_3 = 1200 + 900 + 600 = 2700$.

Problem 3. Find the number of committees of 5 that can be selected from 7 men and 5 women, if the committee is to consist of at least 1 man and at least 1 woman.

Solution.

Men	Women	No. of committees of 5
1	4	${}^7C_1 \times {}^5C_4 = 35$
2	3	${}^7C_2 \times {}^5C_3 = 210$
3	2	${}^7C_3 \times {}^5C_2 = 350$
4	1	${}^7C_4 \times {}^5C_1 = 175$

$\therefore$ Required number of committees containing at least 1 man and at least 1 woman is $35 + 210 + 350 + 175 = 770$.

Problem 4. A woman has 11 close relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following situations: (i) there is no restriction on the choice; (ii) 2 particular persons will not attend separately; (iii) 2 particular persons will not attend together?

Solution. (i) Since there is no restriction on the choice, 5 out of 11 can be invited in ${}^{11}C_5 = 462$ ways.

(ii) Let A and B be the 2 particular persons. The following cases arise:

(a) If both A and B are invited, the number of ways of selection is ${}^9C_3 = 84$.

(b) If both A and B are not invited, the number of ways of selection is ${}^9C_5 = 126$.

$\therefore$ Total number of ways $= 84 + 126 = 210$.

(iii) The following cases arise:

(a) If A is invited and B is not invited: ${}^9C_4 = 126$ ways.

(b) If B is invited and A is not invited: ${}^9C_4 = 126$ ways.

(c) If both A and B are not invited: ${}^9C_5 = 126$ ways.

$\therefore$ Total number of ways $= 126 + 126 + 126 = 378$.

Problem 5. Find the number of arrangements of all the letters of the word TALLAHASSEE. How many of these arrangements have no adjacent A's?

Solution. (i) The given word has 11 letters, in which there are 3 A's; 2 each of L, S, E; and 1 each of T and H.

$$\therefore \text{ Number of such arrangements} = \frac{11!}{3!\,(2!)^3\,(1!)^2} = 8{,}31{,}600.$$

(ii) If the A's are disregarded, the remaining 8 letters can be arranged in $\dfrac{8!}{(2!)^3(1!)^2} = 5040$ ways. In each of these arrangements, there are 9 possible locations for the 3 A's (the gaps marked $\uparrow$):

$$\uparrow \text{T} \uparrow \text{L} \uparrow \text{L} \uparrow \text{H} \uparrow \text{S} \uparrow \text{S} \uparrow \text{E} \uparrow \text{E} \uparrow$$

These locations can be chosen in ${}^9C_3$ ways. $\therefore$ Number of arrangements having no adjacent A's $= 5040 \times {}^9C_3 = 5040 \times 84 = 4{,}23{,}360$.

Problem 6. Find how many distinct 4-digit integers one can make from the digits $1, 3, 3, 7, 7, 8$.

Solution. The following 3 cases arise:

(i) 4-digit integers containing all distinct digits $(1, 3, 7, 8)$: $4! = 24$ ways.

(ii) 4-digit integers containing 2 pairs of identical digits $(3, 3, 7, 7)$: $\dfrac{4!}{2!\,2!} = 6$ ways.

(iii) 4-digit integers containing 1 pair of identical digits: the possible digit sets are

$$\{3,3,1,7\},\ \{3,3,1,8\},\ \{3,3,7,8\},\ \{7,7,1,3\},\ \{7,7,1,8\},\ \{7,7,3,8\},$$

which are 6 in number. This can be done in $\dfrac{4!}{2!\,1!\,1!} \times 6 = 72$ ways.

$\therefore$ Total number of ways $= 24 + 6 + 72 = 102$.

Problem 7. A man has 15 close friends, of whom 6 are women. In how many ways can he invite 3 or more of these friends to a party if he wants the same number of men (including himself) as women?

Solution. Out of the 15 friends, 6 are women and 9 are men.

Women	Men (+ himself in all cases)	No. of ways
2	1	${}^6C_2 \times {}^9C_1 = 135$
3	2	${}^6C_3 \times {}^9C_2 = 720$
4	3	${}^6C_4 \times {}^9C_3 = 1260$
5	4	${}^6C_5 \times {}^9C_4 = 756$
6	5	${}^6C_6 \times {}^9C_5 = 126$

$\therefore$ Total number of ways $= 135+720+1260+756+126 = 2997.$

Problem 8. How many bytes contain (i) exactly two 1's (ii) exactly four 1's (iii) exactly six 1's (iv) at least six 1's?

Solution. 1 byte = 8 bits and each bit is a 0 or 1.
(i) Two 1's can be chosen in ${}^8C_2$ ways and each of the remaining 6 locations is filled by zeros only (1 choice).
$\therefore$ Number of bytes containing exactly two 1's $= {}^8C_2 \times 1^6 = 28.$
(ii) ${}^8C_4 \times 1^4 = 70.$ (iii) ${}^8C_6 \times 1^2 = 28.$
(iv) At least six 1's can be chosen in the following ways: six 1's $\Rightarrow {}^8C_6$; seven 1's $\Rightarrow {}^8C_7$; eight 1's $\Rightarrow {}^8C_8$.
$\therefore$ Number of bytes with at least six 1's is ${}^8C_6 + {}^8C_7 + {}^8C_8 = 28+8+1 = 37.$

Problem 9. Prove the following identities: (i) $C(n,r-1)+C(n,r) = C(n+1,r);$ (ii) $C(m,2)+C(n,2) = C(m+n,2)-mn.$

Solution. (i)

$$\begin{aligned} C(n,r-1)+C(n,r) &= {}^nC_{r-1} + {}^nC_r = \frac{n!}{(n-r+1)!\,(r-1)!} + \frac{n!}{(n-r)!\,r!} \\ &= \frac{n!}{(n-r+1)(n-r)!\,(r-1)!} + \frac{n!}{(n-r)!\,r(r-1)!} \\ &= \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{1}{n-r+1}+\frac{1}{r}\right] = \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{r+n-r+1}{r(n-r+1)}\right] \\ &= \frac{n!}{(n-r)!\,(r-1)!} \times \frac{(n+1)}{r(n-r+1)} = \frac{(n+1)!}{r!\,(n-r+1)!} = C(n+1,r). \end{aligned}$$

(ii)

$$\begin{aligned} C(m,2)+C(n,2) &= \frac{m!}{(m-2)!\,2!} + \frac{n!}{(n-2)!\,2!} = \frac{m(m-1)(m-2)!}{2\,(m-2)!} + \frac{n(n-1)(n-2)!}{2\,(n-2)!} \\ &= \tfrac{1}{2}\left[m(m-1)+n(n-1)\right] = \tfrac{1}{2}\left[m^2+n^2-m-n\right] \\ &= \tfrac{1}{2}\left[(m+n)^2-2mn-m-n\right] = \tfrac{1}{2}\left[(m+n)^2-(m+n)-2mn\right] \\ &= \tfrac{1}{2}\left[(m+n)(m+n-1)\right]-mn = \frac{1}{2}\left[\frac{(m+n)(m+n-1)(m+n-2)!}{(m+n-2)!}\right]-mn \\ &= \frac{1}{2!}\,\frac{(m+n)!}{(m+n-2)!} - mn = C(m+n,2)-mn. \end{aligned}$$

■

3.4 Binomial and Multinomial Theorems

A basic property of $C(n,r)$ is that it is the coefficient of $x^{n-r}y^r$ in the expansion of the expression $(x+y)^n$, where x and y are any two real numbers. In other words, the expansion of $(x+y)^n$ in powers of x and y is as follows:

$$\begin{aligned} (x+y)^n &= x^n + {}^nC_1\,x^{n-1}y + {}^nC_2\,x^{n-2}y^2 + \cdots + {}^nC_{n-1}\,xy^{n-1} + y^n \\ &= \sum_{r=0}^{n} {}^nC_r\,x^{n-r}y^r. \end{aligned} \tag{1}$$

Since ${}^nC_r = {}^nC_{n-r}$,

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\,x^r y^{n-r}. \tag{2}$$

Thus ${}^nC_r$ is the coefficient of $x^r y^{n-r}$ in the expansion of $(x+y)^n$. ${}^nC_r$ is also denoted by $\binom{n}{r}$. Therefore

$$(1) \Rightarrow (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r}y^r, \qquad (2) \Rightarrow (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}. \tag{3}$$

Binomial Theorem

Expression (3) is called the *Binomial Theorem* for a positive integral index n. The numbers $\binom{n}{r}$ for $r = 0, 1, 2, \ldots, n$, namely $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}$, are called the *binomial coefficients*.

The generalization of the Binomial Theorem is known as the Multinomial Theorem.

Multinomial Theorem

For positive integers n and k, the coefficient of $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ in the expansion of $(x_1 + x_2 + \cdots + x_k)^n$ is

$$\frac{n!}{n_1! n_2! \cdots n_k!},$$

where each n_i is a non-negative integer $\leq n$ and $n_1 + n_2 + \cdots + n_k = n$.

Note:

1. The general term in the expansion of $(x_1 + x_2 + \cdots + x_k)^n$ is $\dfrac{n!}{n_1! n_2! \cdots n_k!} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ (alternate statement of the Multinomial Theorem).
2. The expression $\dfrac{n!}{n_1! n_2! \cdots n_k!}$ is also written as $\dbinom{n}{n_1, n_2, \ldots, n_k}$.
3. The Multinomial Theorem can also be stated as

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{n_i} \binom{n}{n_1, n_2, \ldots, n_k} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}.$$

Problems

Problem 1. Prove the following identities for a positive integer n:

$$\text{(i)} \ \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n; \qquad \text{(ii)} \ \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n \binom{n}{n} = 0.$$

Solution. The Binomial Theorem for a positive integral index n is $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$.
(i) Taking $x = 1,\ y = 1$, we get

$$2^n = \sum_{r=0}^{n} \binom{n}{r} 1^r 1^{n-r} = \sum_{r=0}^{n} \binom{n}{r}, \quad \text{i.e.,} \quad 2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}.$$

(ii) Taking $x = -1,\ y = 1$, we get

$$0 = \sum_{r=0}^{n} \binom{n}{r} (-1)^r 1^{n-r} = \sum_{r=0}^{n} (-1)^r \binom{n}{r} \ \Rightarrow \ 0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n \binom{n}{n}.$$

Problem 2. Compute the following: (i) $\dbinom{12}{5,3,2,2}$ (ii) $\dbinom{8}{4,2,2,0}$ (iii) $\dbinom{10}{5,3,2,2}$

Solution. (i) $\dbinom{12}{5,3,2,2} = \dfrac{12!}{5!3!2!2!} = 1{,}66{,}320.$ (ii) $\dbinom{8}{4,2,2,0} = \dfrac{8!}{4!2!2!0!} = 420.$
(iii) Meaningless, since $5+3+2+2 = 12 > n = 10$.

Problem 3. Find the sum of all coefficients in the expansion of (a) $(x+y)^n$ (b) $(x_1 + x_2 + \cdots + x_k)^n$.

Solution. (a) We have $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$. Taking $x=1,\ y=1$, we get

$$2^n = \sum_{r=0}^{n} \binom{n}{r} 1^r 1^{n-r} = \sum_{r=0}^{n} \binom{n}{r}, \quad \text{i.e.,} \quad 2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}.$$

Thus, the sum of all the coefficients in the expansion of $(x+y)^n$ is 2^n.

(b) By the Multinomial Theorem, we have $\sum_{n_i} \binom{n}{n_1, n_2, \ldots, n_k} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} = (x_1 + x_2 + \cdots + x_k)^n$. Taking $x_1 = 1,\ x_2 = 1,\ \ldots,\ x_k = 1$, we get

$$\sum_{n_i} \binom{n}{n_1, n_2, \ldots, n_k} = (\underbrace{1+1+\cdots+1}_{k \text{ times}})^n = k^n.$$

Thus, the sum of all the coefficients in the expansion of $(x_1 + x_2 + \cdots + x_k)^n$ is k^n.

Problem 4. Obtain the sum of all the coefficients in the expansion of (i) $(x+y)^{10}$ (ii) $(x+y+z)^{12}$ (iii) $(x+y+z+w)^5$ (iv) $(2s-3t+5u+6v-11w+3x+2y)^{10}$.

Solution. (i) From result (a) of the above problem, the sum of the coefficients in the expansion of $(x+y)^{10}$ is 2^{10}.
(ii) From result (b) of the above problem, the sum of the coefficients in the expansion of $(x+y+z)^{12}$ is 3^{12}.
(iii) From result (b) again, the sum of the coefficients in the expansion of $(x+y+z+w)^5$ is 4^5.
(iv) By virtue of the Multinomial Theorem, we have

$$\begin{aligned}
&(2s-3t+5u+6v-11w+3x+2y)^{10} \\
&= \sum_{n_i} \binom{10}{n_1, n_2, \ldots, n_7} (2s)^{n_1} (-3t)^{n_2} (5u)^{n_3} (6v)^{n_4} (-11w)^{n_5} (3x)^{n_6} (2y)^{n_7} \\
&= \sum_{n_i} \binom{10}{n_1, n_2, \ldots, n_7} 2^{n_1} (-3)^{n_2} 5^{n_3} 6^{n_4} (-11)^{n_5} 3^{n_6} 2^{n_7} \times s^{n_1} t^{n_2} u^{n_3} v^{n_4} w^{n_5} x^{n_6} y^{n_7}.
\end{aligned}$$

Taking $s = t = u = v = w = x = y = 1$, we get

$$(2-3+5+6-11+3+2)^{10} = \sum_{n_i} \binom{10}{n_1, n_2, \ldots, n_7} 2^{n_1} (-3)^{n_2} 5^{n_3} 6^{n_4} (-11)^{n_5} 3^{n_6} 2^{n_7}.$$

The sum on the RHS of the above expression is the sum of the coefficients in the expansion of $(2s-3t+5u+6v-11w+3x+2y)^{10}$. This sum $= (2-3+5+6-11+3+2)^{10} = 4^{10}$.

Problem 5. Find the coefficient of (i) x^3y^9 in the expansion of $(x+2y)^{12}$; (ii) x^0 in the expansion of $\left(3x^2 - \frac{2}{x}\right)^{15}$; (iii) x^{12} in the expansion of $x^3(1-2x)^{10}$; (iv) x^k in the expansion of $(1+x+x^2)(1+x)^n$, where n is a positive integer and $0 \le k \le n+2$.

Solution. (i) By the Binomial Theorem,

$$(x+2y)^{12} = \sum_{r=0}^{12} \binom{12}{r} x^r (2y)^{12-r} = \sum_{r=0}^{12} \binom{12}{r} 2^{12-r} x^r y^{12-r}.$$

Taking $r = 3$ in the above expansion, the coefficient of x^3y^9 is $\binom{12}{3} 2^9 = \frac{12!}{9!3!} \cdot 2^9 = 1{,}12{,}640$.

(ii) By the Binomial Theorem,

$$\begin{aligned}
\left(3x^2 - \frac{2}{x}\right)^{15} &= \sum_{r=0}^{15} \binom{15}{r} (3x^2)^r \left(-\frac{2}{x}\right)^{15-r} = \sum_{r=0}^{15} \binom{15}{r} 3^r (-2)^{15-r} x^{2r} \left(\frac{1}{x}\right)^{15-r} \\
&= \sum_{r=0}^{15} \binom{15}{r} 3^r (-2)^{15-r} x^{2r} x^{r-15} = \sum_{r=0}^{15} \binom{15}{r} 3^r (-2)^{15-r} x^{3r-15}.
\end{aligned}$$

Taking $r = 5$ in the above expansion $(3r - 15 = 0)$, the coefficient of x^0 is

$$\binom{15}{5} 3^5 (-2)^{10} = {}^{15}C_5 \cdot 3^5 \cdot (-2)^{10} = 74{,}72{,}42{,}496.$$

(iii) By the Binomial Theorem,

$$(1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} 1^{10-r} (-2x)^r \ \therefore\ x^3(1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} (-2)^r x^{r+3}.$$

Taking $r = 9$ in the above expansion, we get the coefficient of x^{12} as $\binom{10}{9}(-2)^9 = {}^{10}C_9 \times (-2)^9 = -5{,}120$.

(iv) Using the Binomial Theorem, $(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} 1^{n-r} x^r$.

$$\therefore\ (1+x+x^2)(1+x)^n = (1+x+x^2) \sum_{r=0}^{n} \binom{n}{r} x^r = \sum_{r=0}^{n} \binom{n}{r} x^r + \sum_{r=0}^{n} \binom{n}{r} x^{r+1} + \sum_{r=0}^{n} \binom{n}{r} x^{r+2}.$$

The coefficient of x^k is obtained by taking $r = k$ in the 1st sum, $r = k-1$ in the 2nd sum and $r = k-2$ in the 3rd sum, i.e., the coefficient of x^k is

$$\binom{n}{k} + \binom{n}{k-1} + \binom{n}{k-2}.$$

For example, the coefficient of x^{15} is $\binom{n}{15} + \binom{n}{14} + \binom{n}{13}$.

Problem 6. Determine the coefficient of (i) xyz^5 in the expansion of $(x+y+z)^7$; (ii) $x^2y^2z^3$ in the expansion of $(3x-2y-4z)^7$; (iii) $x^{11}y^4z^2$ in the expansion of $(2x^3 - 3xy^2 + z^2)^6$; (iv) xyz^{-2} in the expansion of $(x - 2y + 3z^{-1})^4$; (v) $a^2b^3c^2d^5$ in the expansion of $(a+2b-3c+2d+5)^{16}$.

Solution. (i) By the Multinomial Theorem, the general term in the expansion of $(x+y+z)^7$ is

$$\binom{7}{n_1, n_2, n_3} x^{n_1} y^{n_2} z^{n_3}.$$

For $n_1 = 1,\ n_2 = 1,\ n_3 = 5$, we have

$$\binom{7}{1,1,5} xyz^5 = \frac{7!}{1!1!5!} xyz^5 = 42xyz^5.$$

This shows that the required coefficient of xyz^5 is 42.

(ii) The general term in the expansion of $(3x-2y-4z)^7$ is $\binom{7}{n_1, n_2, n_3} (3x)^{n_1} (-2y)^{n_2} (-4z)^{n_3}$. For $n_1 = 2,\ n_2 = 2,\ n_3 = 3$, we have

$$\binom{7}{2,2,3} (3x)^2 (-2y)^2 (-4z)^3 = \binom{7}{2,2,3} 3^2 (-2)^2 (-4)^3 x^2 y^2 z^3.$$

$\therefore$ Coefficient of $x^2y^2z^3$ is $\binom{7}{2,2,3} \times 9 \times 4 \times (-64) = -2304 \times \frac{7!}{2!2!3!} = -4{,}83{,}840.$

(iii) The general term in the expansion of $(2x^3 - 3xy^2 + z^2)^6$ is

$$\begin{aligned} \binom{6}{n_1, n_2, n_3} (2x^3)^{n_1} (-3xy^2)^{n_2} (z^2)^{n_3} &= \binom{6}{n_1, n_2, n_3} 2^{n_1} (-3)^{n_2} x^{3n_1} x^{n_2} y^{2n_2} z^{2n_3} \\ &= \binom{6}{n_1, n_2, n_3} 2^{n_1} (-3)^{n_2} x^{3n_1 + n_2} y^{2n_2} z^{2n_3}. \end{aligned}$$

For $x^{11}y^4z^2$: $2n_2 = 4 \Rightarrow n_2 = 2$; $3n_1 + 2 = 11 \Rightarrow n_1 = 3$; $2n_3 = 2 \Rightarrow n_3 = 1$ (and indeed $3+2+1 = 6$). We have

$$\binom{6}{3,2,1} 2^3 (-3)^2 x^{11} y^4 z^2.$$

$\therefore$ Coefficient of $x^{11}y^4z^2$ is $2^3(-3)^2 \binom{6}{3,2,1} = 8 \times 9 \times \dfrac{6!}{3!\,2!\,1!} = 4{,}320.$

(iv) The general term in the expansion of $(x - 2y + 3z^{-1})^4$ is

$$\binom{4}{n_1, n_2, n_3} x^{n_1} (-2y)^{n_2} (3z^{-1})^{n_3} = \binom{4}{n_1, n_2, n_3} (-2)^{n_2} 3^{n_3} x^{n_1} y^{n_2} z^{-n_3}.$$

For $n_1 = 1,\ n_2 = 1,\ n_3 = 2$, we have $\binom{4}{1,1,2} (-2)^1 \cdot 3^2 xyz^{-2}$. $\therefore$ Coefficient of xyz^{-2} is $-2 \cdot 9 \times \binom{4}{1,1,2} = -18 \times \dfrac{4!}{1!\,1!\,2!} = -216.$

(v) The general term in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$ is

$$\binom{16}{n_1, n_2, n_3, n_4, n_5} a^{n_1} (2b)^{n_2} (-3c)^{n_3} (2d)^{n_4} 5^{n_5}.$$

For $n_1 = 2,\ n_2 = 3,\ n_3 = 2,\ n_4 = 5$: $n_5 = 16 - (2+3+2+5) = 4$ [$\because n_1 + \cdots + n_5 = n \Rightarrow n_5 = n - (n_1 + \cdots + n_4)$], we have

$$\binom{16}{2,3,2,5,4} 2^3 \cdot (-3)^2 \cdot 2^5 \cdot 5^4 a^2 b^3 c^2 d^5.$$

$\therefore$ Coefficient of $a^2b^3c^2d^5$ is

$$\begin{aligned} 8 \times 9 \times 2^5 \times 5^4 \times \binom{16}{2,3,2,5,4} &= 14{,}40{,}000 \times \frac{16!}{2!\,3!\,2!\,5!\,4!} \\ &= 14{,}40{,}000 \times 30{,}27{,}02{,}400 = 4{,}35{,}89{,}14{,}56{,}00{,}000. \end{aligned}$$

Homework

Find the coefficient of:

1. x^5y^2 in the expansion of $(2x - 3y)^7$.
2. x^3z^4 in the expansion of $(x + y + z)^7$.
3. $x^3y^3z^2$ in the expansion of $(2x - 3y + 5z)^8$.
4. $w^3x^2yz^2$ in the expansion of $(2w - x + 3y - 2z)^8$.
5. $x_1^2x_3x_4^3x_5^4$ in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$.

3.5 Combinations with Repetitions

Suppose we have to select a combination of r objects, with repetitions, from a set of n distinct objects. Then the number of such selections is given by

$$C(n + r - 1, r) = \frac{(n + r - 1)!}{r!\,(n-1)!} = C(r + n - 1, n - 1).$$

i.e.,

$$\boxed{C(n + r - 1, r) = C(r + n - 1, n - 1)} \tag{1}$$

represents the number of combinations of n distinct objects, taken r at a time, with repetitions allowed.

Note:

1. Eq. (1) represents the number of ways in which r identical objects can be distributed among n distinct containers (each container may receive $0,1,2,3,\ldots$ objects).
2. Eq. (1) also represents the number of non-negative integer solutions of the equation $x_1+x_2+\cdots+x_n=r$, i.e., with $x_i\geq 0$ for $i=1,2,\ldots,n$.
   [A non-negative integer solution of $x_1+x_2+\cdots+x_n=r$ is an n-tuple $(x_1,x_2,\ldots,x_n)$, where $x_1,x_2,\ldots,x_n$ are non-negative integers whose sum is r.]
3. *How to identify n and r:* In an ordinary combination $(n\geq r)$ we select r out of n. In a combination with repetition, r may be greater than n: n is the number of distinct places (containers / variables) and r is the number of identical items to be distributed (the sum to be reached).

Problems

Problem 1. In how many ways can we distribute 10 identical marbles among 6 distinct containers?

Solution. Here $r=10,\ n=6$. $\therefore$ Required number $=C(6+10-1,10)=C(15,10)=\dfrac{15!}{10!\,5!}=3003$.

Problem 2. Find the number of ways of placing 8 identical balls in 5 numbered boxes.

Solution. Here $r=8,\ n=5$. Required number $=C(5+8-1,8)=C(12,8)=\dfrac{12!}{8!\,4!}=495$.

Problem 3. A bag contains coins of 7 different denominations, with at least 1 dozen coins in each denomination. In how many ways can we select a dozen coins from the bag?

Solution. Here $r=12,\ n=7$. The number of ways of making a selection of a dozen coins from the bag is

$$C(7+12-1,12)=C(18,12)=\frac{18!}{12!\,6!}=18{,}564.$$

Problem 4. Find the number of non-negative integer solutions of the equation $x_1+x_2+x_3+x_4+x_5=8$.

Solution. Here $r=8,\ n=5$. The required number is $C(5+8-1,8)=C(12,8)=\dfrac{12!}{8!\,4!}=495$.

Problem 5. Find the number of non-negative integer solutions of the equation $x_1+x_2+x_3+x_4=7$.

Solution. Here $n=4,\ r=7$. Required number $=C(4+7-1,7)=C(10,7)=\dfrac{10!}{7!\,3!}=120$.

Problem 6. Find the number of distinct terms in the expansion of $(x_1+x_2+x_3+x_4+x_5)^{16}$.

Solution. The general term in the expansion of $(x_1+x_2+x_3+x_4+x_5)^{16}$ is of the form

$$\binom{16}{n_1,n_2,n_3,n_4,n_5}x_1^{n_1}x_2^{n_2}x_3^{n_3}x_4^{n_4}x_5^{n_5},$$

where each n_i is a non-negative integer and the sum of these n_i's is 16. $\therefore$ Number of distinct terms in the expansion $=$ number of non-negative integer solutions of $n_1+n_2+n_3+n_4+n_5=16$. $\therefore$ Required number $=C(5+16-1,16)=C(20,16)=4845$.

Problem 7. Find the number of non-negative integer solutions of the inequality $x_1+x_2+\cdots+x_6<10$.

Solution. Given $x_1+x_2+\cdots+x_6<10$, i.e., $x_1+x_2+\cdots+x_6\leq 9$. Write

$$x_1+x_2+\cdots+x_6=9-x_7,$$

so that x_7 is a non-negative integer (e.g. $2+3<10 \Rightarrow 2+3=9-4$). $\therefore$ The required number is the number of non-negative solutions of the equation $x_1+x_2+\cdots+x_7=9$. This number is $C(7+9-1,9)=C(15,9)=5005$.

Problem 8. Find the number of distinct terms in the expansion of $(w+x+y+z)^{10}$.

Solution. The general term in the expansion of $(w+x+y+z)^{10}$ is $\binom{10}{n_1,n_2,n_3,n_4} w^{n_1}x^{n_2}y^{n_3}z^{n_4}$, where each n_i is a non-negative integer and the sum of these n_i's is 10. $\therefore$ Number of distinct terms in the expansion = number of non-negative integer solutions of $n_1+n_2+n_3+n_4=10$. This number $=C(4+10-1,10)=C(13,10)=286$.

Problem 9. Find the number of non-negative integer solutions of the inequality $x_1+x_2+\cdots+x_5 \le 19$.

Solution. Given $x_1+x_2+\cdots+x_5 \le 19 \Rightarrow x_1+x_2+\cdots+x_5=19-x_6$, so that x_6 is non-negative. $\therefore$ Required number = number of non-negative integer solutions of the equation $x_1+x_2+\cdots+x_5+x_6=19$. This number is $C(6+19-1,19)=C(24,19)=42{,}504$.

Problem 10. Find the number of integer solutions of

$$x_1+x_2+\cdots+x_5=30, \quad \text{where } x_1\ge 2,\ x_2\ge 3,\ x_3\ge 4,\ x_4\ge 2,\ x_5\ge 0. \tag{1}$$

Solution. Let $y_1=x_1-2;\ y_2=x_2-3;\ y_3=x_3-4;\ y_4=x_4-2;\ y_5=x_5$. Then $y_1,y_2,\ldots,y_5$ are all non-negative integers ($y_i\ge 0$ for $i=1,2,\ldots,5$). The given equation now becomes

$$\begin{aligned}&(y_1+2)+(y_2+3)+(y_3+4)+(y_4+2)+y_5=30\\ \Rightarrow\ &y_1+y_2+y_3+y_4+y_5=30-11=19.\end{aligned} \tag{2}$$

Number of non-negative integer solutions of (2) is $C(5+19-1,19)=C(23,19)=8855$.

Problem 11. How many integer solutions are there to $x_1+x_2+\cdots+x_5=20$, where $x_i\ge 2$?

Solution. Given $x_1\ge 2,\ x_2\ge 2,\ \ldots,\ x_5\ge 2$. Let $y_1=x_1-2,\ y_2=x_2-2,\ \ldots,\ y_5=x_5-2$. $\therefore$ The given equation becomes

$$(y_1+2)+(y_2+2)+(y_3+2)+(y_4+2)+(y_5+2)=20 \ \Rightarrow\ y_1+y_2+\cdots+y_5=10.$$

$\therefore$ Required number $=C(5+10-1,10)=C(14,10)=1001$.

Problem 12. In how many ways can we distribute 12 identical pencils to 5 children so that every child gets at least 1 pencil?

Solution. First, let us distribute one pencil to each child. The remaining 7 pencils are to be distributed to 5 children ($n=5,\ r=7$). This can be done in

$$C(5+7-1,7)=C(11,7)=330 \text{ ways.}$$

Problem 13. In how many ways can 10 identical pens be distributed among 5 children in the following cases: (i) there are no restrictions; (ii) each child gets at least one pen; (iii) the youngest child gets at least 2 pens?

Solution. (i) Since there are no restrictions in distributing 10 pens to 5 children, the number of ways of distribution is $C(5+10-1,10)=C(14,10)=1001$.

(ii) First we distribute one pen to each child. Then the remaining 5 pens are to be distributed among 5 children in $C(5+5-1,5)=C(9,5)=126$ ways.

(iii) First we give 2 pens to the youngest child; then the remaining 8 pens are to be distributed among 5 children. This can be done in $C(5+8-1,8)=C(12,8)=495$ ways.

Problem 14. In how many ways can one distribute eight identical balls into 4 distinct containers so that (i) no container is empty? (ii) the 4th container gets an odd number of balls?

Solution. (i) First we distribute one ball to each container. Then the remaining 4 balls are to be distributed among 4 containers, and this can be done in $C(4+4-1,4) = C(7,4) = 35$ ways.

(ii) If the 4th container has to get an odd number of balls, we have to put 1 or 3 or 5 or 7 balls into it.
Case (i): Suppose we put 1 ball in the 4th container; then the remaining 7 balls can be distributed into the remaining 3 containers in $C(3+7-1,7) = C(9,7)$ ways.
Case (ii): Suppose we put 3 balls in the 4th container; then the remaining 5 balls can be distributed into the remaining 3 containers in $C(3+5-1,5) = C(7,5)$ ways.
Case (iii): Suppose we put 5 balls in the 4th container; then the remaining 3 balls can be distributed into the remaining 3 containers in $C(3+3-1,3) = C(5,3)$ ways.
Case (iv): Suppose we put 7 balls in the 4th container; then the remaining 1 ball can be distributed in $C(3+1-1,1) = C(3,1)$ ways.
$\therefore$ Total number of ways of distributing the given balls with the given condition is

$$C(9,7)+C(7,5)+C(5,3)+C(3,1) = 36+21+10+3 = 70.$$

Problem 15. Find the number of ways of distributing 7 identical pens and 7 identical pencils to 5 children so that each gets at least 1 pen and at least 1 pencil.

Solution. First, let us distribute 1 pen and 1 pencil to each child. Then the remaining 2 pens can be distributed to 5 children in $C(5+2-1,2) = C(6,2)$ ways. Similarly, the remaining 2 pencils can be distributed to 5 children in $C(5+2-1,2) = C(6,2)$ ways. $\therefore$ The number of ways of distributing the given sets under the given condition is

$$C(6,2) \times C(6,2) = 15 \times 15 = 225.$$

Problem 16. Find the number of ways of giving 10 identical gift boxes to 6 persons A, B, C, D, E, F in such a way that the total number of boxes given to A and B together does not exceed 4.

Solution. Of the 10 boxes, suppose r boxes are given to A and B together; then $0 \le r \le 4$. $\therefore$ Number of ways of giving r boxes to A and B is

$$C(2+r-1,r) = C(r+1,r) = \frac{(r+1)!}{r!\,1!} = \frac{(r+1)\,r!}{r!} = r+1. \tag{1}$$

Number of ways in which the remaining $(10-r)$ boxes can be given to C, D, E, F is

$$C\big(4+(10-r)-1,\,(10-r)\big) = C(13-r,\,10-r). \tag{2}$$

Consequently, the number of ways in which r boxes may be given to A and B and $(10-r)$ boxes to C, D, E, F is $(r+1) \times C(13-r,10-r)$. Since $0 \le r \le 4$, the total number of ways in which the boxes may be given is

$$\begin{aligned}\sum_{r=0}^{4}(r+1) \times C(13-r,\,10-r) &= 1(286)+2(220)+3(165)+4(120)+5(84)\\ &= 286+440+495+480+420 = 2121.\end{aligned}$$

Problem 17. A total amount of Rs 1500 is to be distributed to 3 poor students A, B, C of a class. In how many ways can the distribution be made in multiples of Rs 100 if (i) every one of these must get at least Rs 300? (ii) A must get at least Rs 500, and B and C must get at least Rs 400 each?

Solution. Taking Rs 100 as a unit, there are 15 units for distribution among 3 students A, B, C.
Case (i): Each of the 3 students must get at least 3 units. So let us first distribute 3 units to each of A, B, C;

then the remaining 6 units are to be distributed to A, B, C, and this can be done in $C(3+6-1,6) = C(8,6) = 28$ ways.
Case (ii): A must get at least 5 units, and B and C must get at least 4 units each. So let us first distribute 5 units to A and 4 units each to B and C; then the remaining 2 units are to be distributed to A, B, C. This can be done in $C(3+2-1,2) = C(4,2) = 6$ ways.

Problem 18. A total of Rs 10,000 is to be distributed to 4 persons A, B, C, D in multiples of Rs 1000. In how many ways can the distribution be done (i) if there is no restriction? (ii) if every one of these persons should receive at least Rs 1000? (iii) if every one should receive at least Rs 1000 and A in particular should receive at least Rs 5000?

Solution. Taking Rs 1000 as one unit, there are 10 units for distribution among 4 persons.
Case (i): Since there is no restriction, 10 units can be distributed among 4 persons in $C(4+10-1,10) = C(13,10) = 286$ ways.
Case (ii): Each of the 4 persons must get at least 1 unit. Let us first distribute 1 unit to each of A, B, C, D; then the remaining 6 units are to be distributed among 4 persons. This can be done in $C(4+6-1,6) = C(9,6) = 84$ ways.
Case (iii): A should receive at least 5 units and each of B, C, D should receive at least 1 unit. Let us first distribute 5 units to A and 1 unit each to B, C, D; then the remaining 2 units are to be distributed among 4 persons. This can be done in $C(4+2-1,2) = C(5,2) = 10$ ways.

Problem 19. A message is made up of 12 different symbols and is to be transmitted through a communication channel. In addition to the 12 symbols, the transmitter will also send a total of 45 blank spaces between the symbols, with at least 3 spaces between each pair of consecutive symbols. In how many ways can the transmitter send such a message?

Solution. The 12 symbols can be arranged in 12! ways. For each of these arrangements, there are 11 positions between the 12 symbols. Since there must be at least 3 spaces between consecutive symbols, we use up 33 blank spaces out of 45. The remaining 12 $(= 45-33)$ blank spaces can be accommodated in the 11 positions in $C(11+12-1,12) = C(22,12)$ ways. $\therefore$ Number of ways the transmitter can send such a message is

$$C(22,12) \times 12! = \frac{22!}{10!\,12!} \times 12! = \frac{22!}{10!} \approx 3.0974 \times 10^{14}.$$

Problem 20. Six distinct symbols are transmitted through a communication channel. A total of 12 blanks are to be inserted between the symbols, with at least 2 blanks between every pair of symbols. In how many ways can the symbols and the blanks be arranged?

Solution. The 6 symbols can be arranged in 6! ways. For each of these arrangements, there are 5 positions between the 6 symbols. Since there must be at least 2 blanks between every pair of symbols, 10 of the 12 blanks will be used up. The remaining 2 blanks are to be accommodated in the 5 positions. This can be done in $C(5+2-1,2) = C(6,2)$ ways. $\therefore$ Number of ways the symbols and the blanks can be arranged is $C(6,2) \times 6! = 15 \times 720 = 10{,}800$.

End of Module 2